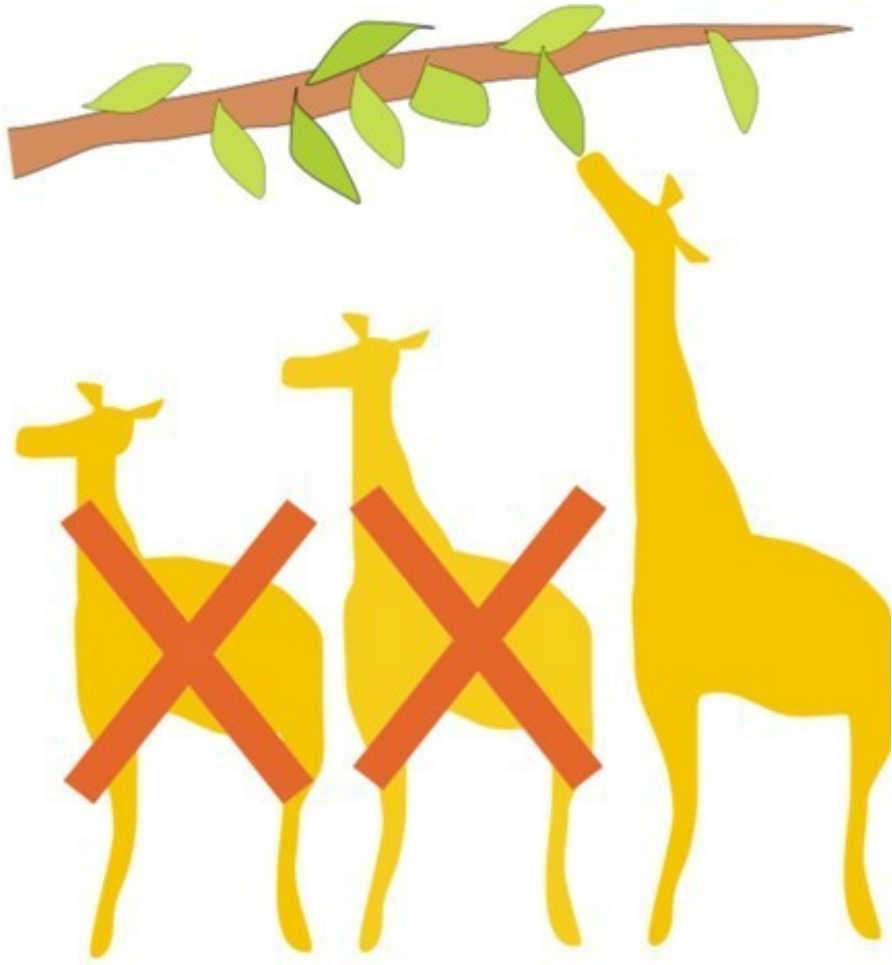


Evolution by Natural Selection



Natural Selection as an Algorithm

- **Variation:** individuals in a population vary in their traits; they have different **genotypes** (sets of genes) leading to different **phenotypes** (differences in physical bodies and behaviors).
- **Inheritance:** those traits are passed on to offspring
- **Selection:** traits that help organisms survive and reproduce are “selected”; they become more common over generations. Sometimes also called “**differential survival and reproduction**”.

Evolution by Natural Selection



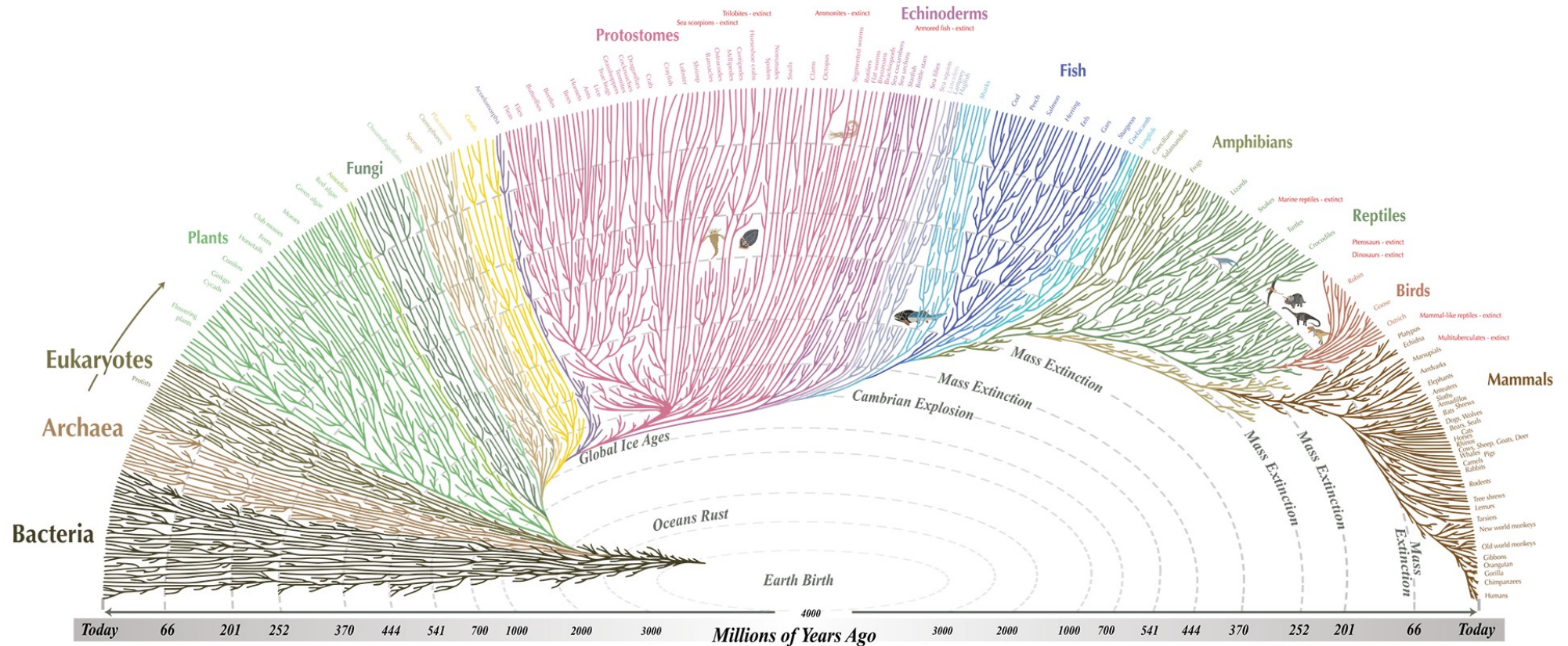
Example

- Peppered moths (light vs. dark)
- Industrial Revolution → pollution darkened trees
- Selection favored darker moths

Evolution by Natural Selection

Evolution ≠ Progress

- Not “progress” or a ladder
- Evolution = adaptation to environment
- Traits can be gained or *lost*, depending on whether those traits benefit survival or reproduction in the current environment.



All the major and many of the minor living branches of life are shown on this diagram, but only a few of those that have gone extinct are shown. Example: Dinosaurs - extinct



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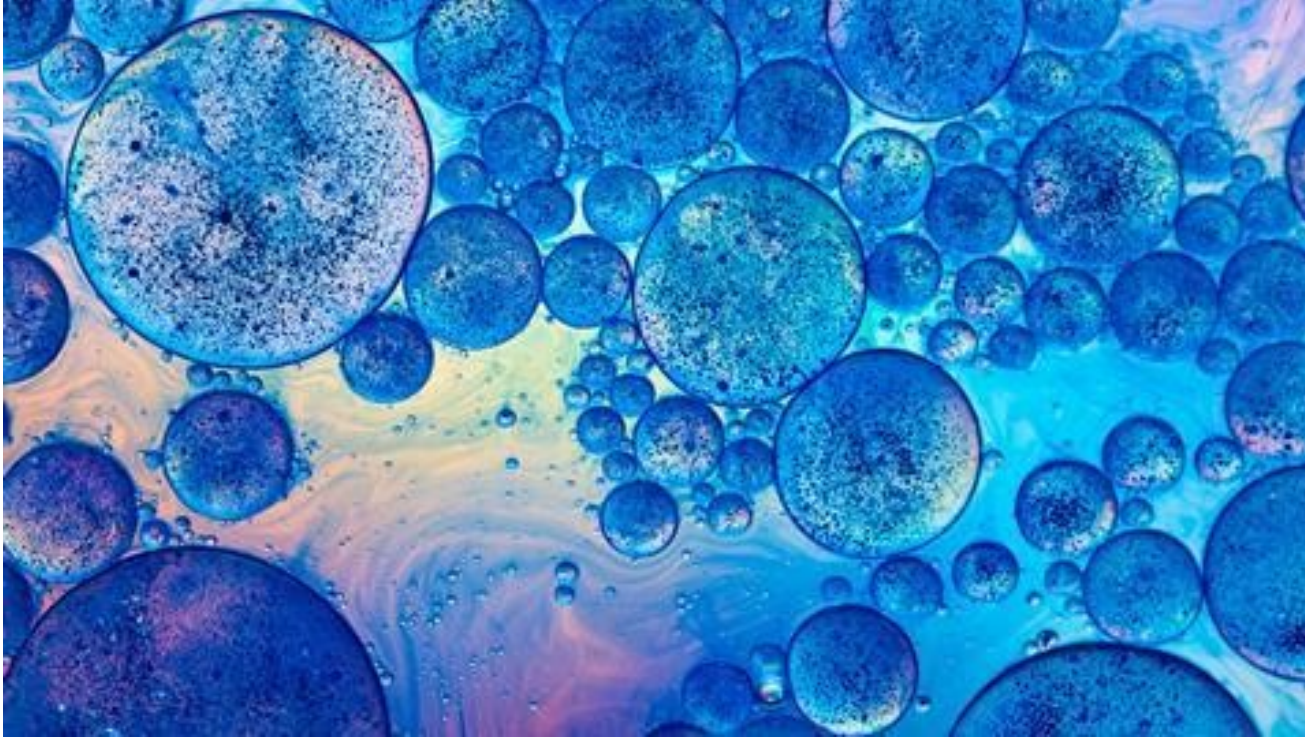
The Evolution of the First Cells

Early Earth, Abiogenesis, & RNA World Hypothesis



- ~4 billion years ago: early Earth
- random chemistry resulted in short RNA chains
- Some RNA happened to be ribozymes (catalysts)
- Rare ribozymes helped copying → **replication**
- Copying was imperfect → **variation**
- Some variants were better at surviving or copying themselves → **selection**
- Variation + inheritance + selection = evolution begins

The Evolution of the First Cells



Early Adaptations

- Lipids resulting in membranes (protect & contain RNA)
- Ribozymes develop into ribosomes (protein factories)
- Proteins can enrich complexity: catalysts, energy metabolizers, motors, sensors
- DNA = stable blueprint

The Evolution of the First Cells



How do we know this?

- chemistry shows it's possible,
- biology shows traces of an RNA world still remain,
- geology sets the timeline,
- and comparative genomics shows what early cells must have had.

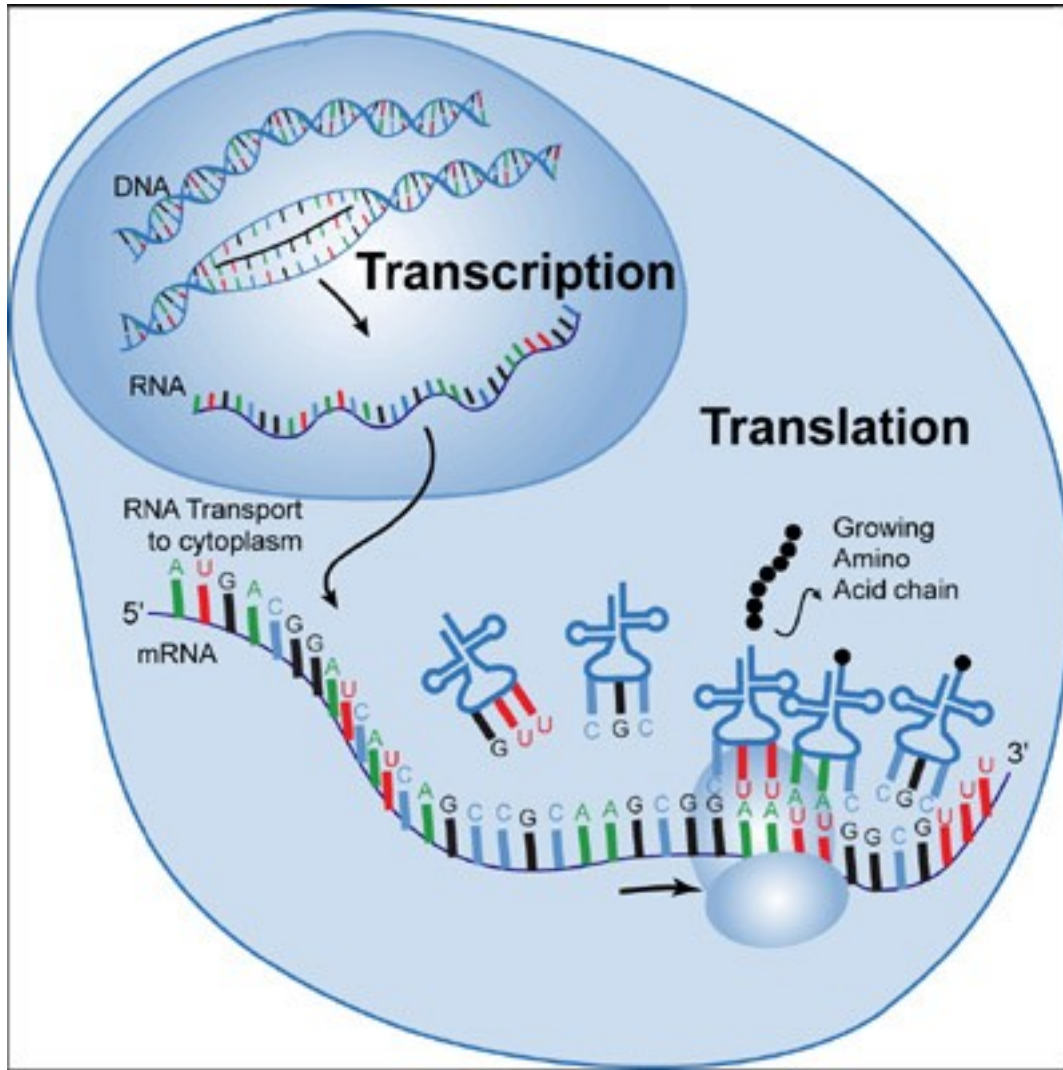
The Evolution of the First Cells



Early Intelligence in Single-Celled Life

- Single cells with proteins for:
 - Movement (propellers, motors)
 - Sensing (light, temperature, chemicals)
- This resulted in behavior:
 - Move toward food
 - Avoid danger
- Cells with adaptive behaviors had a better chance of survival and reproduction, and so came to outnumber those that did not.

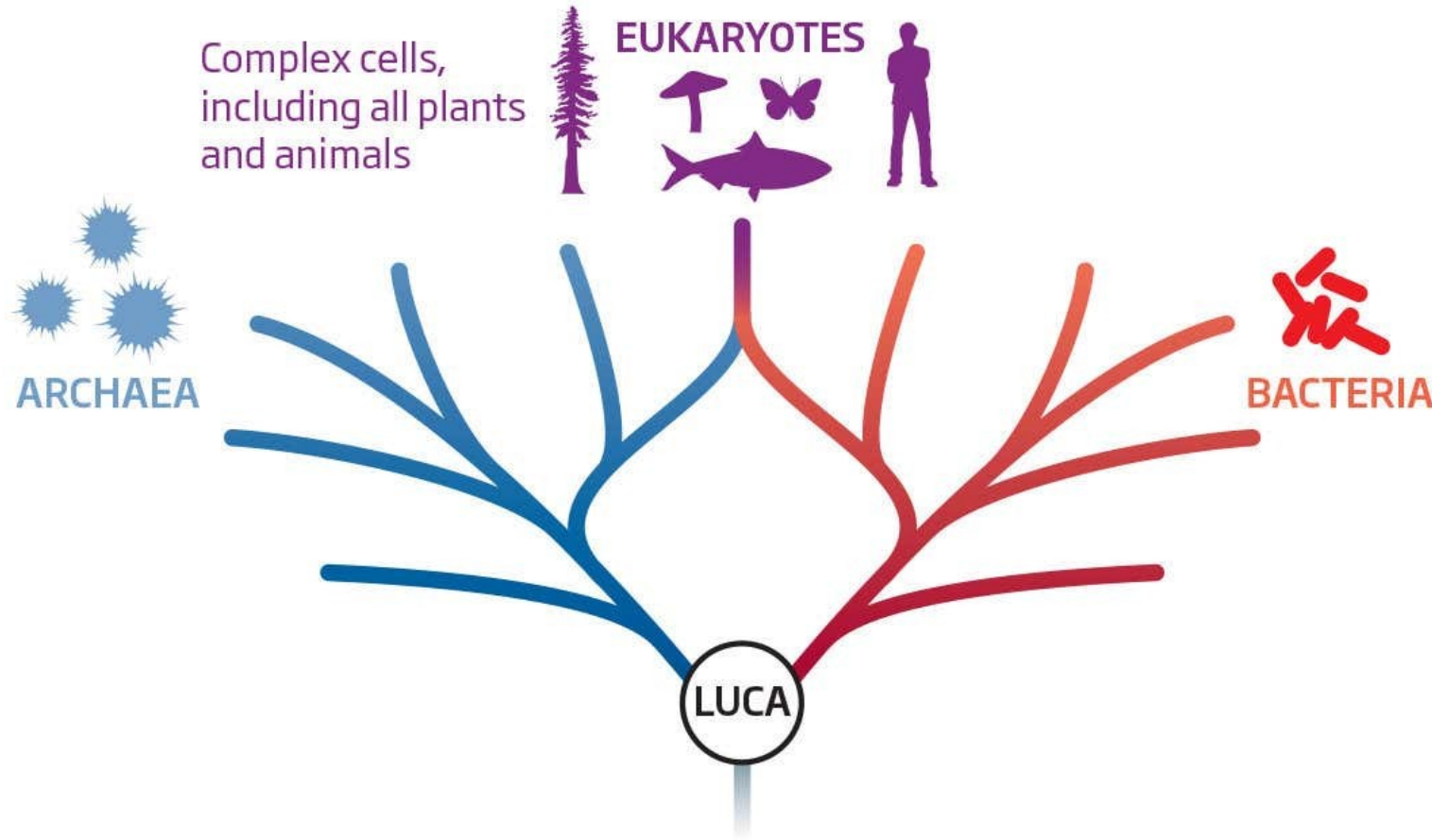
The Evolution of the First Cells



DNA as Information Storage

- DNA = stable blueprint
- RNA = messenger, helper
- Proteins = machines
- Central Dogma: DNA → RNA → Protein

The Evolution of the First Cells



LUCA: Last Universal Common Ancestor

- ~3.5 billion years ago
- Shared by all life today
- Features:
 - DNA, RNA, proteins
 - Lipid membranes
 - ATP metabolism